



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	MATHEMATICAL PHYSICS				
Type of Course	Core in Major				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Fundamentals of vectors, calculus and kinematics.				
Course Summary	This course explores Newton's Laws of Motion and how they can be applied to solve different mechanical systems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate proficiency in manipulating matrices and tensors algebraically and geometrically	Applying	Procedural Knowledge	Written exams, problem sets
CO2	Apply various transforms such as Fourier, Laplace, and Z-transforms to analyze signals and systems	Applying	Procedural Knowledge	Homework assignments, exams

CO3	Understand the properties and applications of special functions such as Bessel, Legendre, and Hermite functions	Understanding	Conceptual Knowledge	Class discussions, presentations
CO4	Solve differential equations using series solutions methods, including power series and Frobenius methods	Applying	Procedural Knowledge	Laboratory experiments, simulations
CO5	Analyze the behavior of complex functions, including their mappings and singularities, in the complex plane	Analyzing	Conceptual Knowledge	Projects, research papers
CO6	Utilize complex analysis techniques to solve problems in physics, engineering, and other applied fields	Applying	Procedural Knowledge	Design projects, presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	MATRICES AND TENSORS		11	17
	1	Linear Combinations, Linear Functions, Linear Operators; Linear Dependence and Independence	2	
	2	Special Matrices and Formulas	1	
	3	Eigenvalues and Eigenvectors; Diagonalizing Matrices	2	
	4	Tensors: Introduction, Cartesian Tensors, Tensor Notation and Operations; Kronecker Delta and Levi-Civita Symbol, Pseudovectors and Pseudotensors	3	
	5	Curvilinear Coordinates and Metric Tensor; Vector operators in Orthogonal Curvilinear Coordinates	2	
	6	Non-cartesian Tensors	1	

	Sections 3.7 – 3.9, 3.11 of chapter 3, and sections 10.1 – 10.3, 10.5, 10.6, 10.8 – 10.10 of chapter 10 of Book 1 Self-study: Sections 3.1 – 3.6, 3.10, 3.12, 10.4, 10.7 of Book 1			
II	FOURIER TRANSFORMS AND SPECIAL FUNCTIONS		11	17
	7	Fourier Coefficients; Dirichlet Conditions; Complex Form of Fourier Series; Other Intervals	3	
	8	Fourier Series of Even and Odd Functions; Parseval's Theorem	1	
	9	Fourier Transforms, Parseval's Theorem for Fourier Integrals	2	
	10	Dirac Delta Function	2	
	11	Factorial Function; Gamma Function; Recursion Relation	1	
	12	Gamma Function of Negative Numbers; Some Important Formulas Involving Gamma Functions; Stirling's Formula	2	
	Sections 7.5 – 7.9, 7.11, 7.12 of chapter 7, sections 11.2 – 11.5, 11.11 of chapter 11 of Book 1 Self-study: Sections 7.1 – 7.4, 7.10, 8.8 – 8.11, 11.6 – 11.9 of Book 1			
III	SERIES SOLUTIONS OF DIFFERENTIAL EQUATIONS		13	20
	13	Introduction; Legendre's Equation; Legendre Polynomials; Rodrigues' Formula	3	
	14	Generating Function for Legendre Polynomials; Recursion Relations	2	
	15	Orthogonality and Normalization of Legendre Polynomials; Legendre Series	2	
	16	Associated Legendre Functions	1	
	17	Generalized Power Series (Method of Frobenius)	1	
	18	Bessel's Equation and Its Second Solution; Graphs and Zeros of Bessel Functions; Recursion Relations	3	
	19	Orthogonality of Bessel Functions	1	
	Sections 12.1, 12.2, 12.4, 12.5, 12.7 – 12.15, 12.19 of chapter 12 Book 1			

	Self-study: Sections 12.3, 12.6, 12.16, 12.17, 12.22 of Book 1			
IV	COMPLEX FUNCTIONS		10	16
	20	Introduction; Analytic Functions	2	
	21	Contour Integrals	2	
	22	Laurent Series	1	
	23	Residue Theorem; Methods of Finding Residues	2	
	24	Evaluation of Definite Integrals by the Use of Residue Theorem; Residues at Infinity	3	
	Sections 14.1 – 14.8 of chapter 14 of Book 1 Self-study: section 14.9 and 14.10 of Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Using matrix inversion, solve the system of homogeneous linear equations.		
	2	Simulate and verify that the trace is unchanged after diagonalization.		
	3	Simulate the square wave, triangular wave and sawtooth wave using Fourier series (See section 5.12 of Book 2).		
	4	Simulate and analyzing periodic signals using Fourier transform.		
	5	Simulate the Bessel and Spherical Bessel functions.		
	6	Simulate the Legendre and Associated Legendre functions and Spherical harmonics.		
	7	Simulate the Laguerre and Associated Laguerre Polynomials.		
	8	Simulate the Hermite Polynomials.		

	9	Simulate the Airy Functions.		
	10	Simulate and solve differential equations by power series method.		

Books and References:

1. Mathematical Methods in the Physical Sciences (3rd Edition, Indian Adaptation, Wiley) by Mary L Boas (Book 1)
2. Mathematical Methods for Physics and Engineering by K F Riley, M P Hobson and S J Bence, 3rd edition
3. Mathematical Methods for Physicists by G.B Arfken and H J Weber (Academic Press)
4. Advanced Engineering Mathematics by Erwin Kreyzig (Wiley)
5. NPTEL video lectures: <https://nptel.ac.in/courses/115106086>
6. Python for Education by Dr. B P Ajithkumar, IUAC, New Delhi; e-book freely downloadable from <https://scischool.in/python/index.html>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	0	1	0	2	1	1	0	1	1	1	1	0
CO 2	0	0	3	0	2	1	1	0	1	1	1	1	0
CO 3	1	0	2	0	3	1	1	0	1	1	1	1	0
CO 4	0	1	3	0	2	1	1	0	1	1	1	1	0
CO 5	0	0	2	0	3	2	1	0	1	1	2	1	0
CO 6	1	0	3	1	2	2	1	0	1	1	2	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	